



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition, to part I (General Handout for all courses appended to the Time-Table), this portion gives further specific details regarding the course.

Course No : EEE G595
Course Title : NANO-ELECTRONICS AND NANO-PHOTONICS TECHNOLOGY
Instructor-in-Charge : V K CHAUBEY

1. Course Description:

Semiconductor fundamentals for Nanotechnology, Introduction to low dimensional nano-structures and Quantum Mechanics, Scaling and nanoelectronics, Nano electronic device materials and circuits, Strain engineering and high speed quantum electronic devices, Optical properties of nano-structures, Linear and non linear effects of optical materials and structures, Nano photonic devices, Electronic Devices based on nano structures in evolution of Silicon Base Devices, Integrated photo electronic devices and systems and its applications, New Trends in Nano devices and technology.

2. Scope and Objective of the Course:

This course provides a blend of electronic and photonic fundamentals and their applications to the fast moving technology involved in high-speed electronics, optical electronics and optical communication systems in terms of nano structures. Electronics and Photonics are becoming increasingly important as limitations of speed, size and bandwidth affect many electronic devices and systems. Moreover, integration of electronics and photonics becomes an important issue related to modern technical aspects like optical interconnects, integrated optical circuits etc. Here the fundamentals of semiconductors, quantum structures and transport properties related to quantum structures will be first discussed. That will be followed by electronic and optical properties and details of strain engineering. Finally all these basic concepts will be applied in understanding modern and photonic and electronic devices useful for engineering applications. Students will also be encouraged to learn hands-on-training in analysis/simulation of some electronic-photonic devices using MATLAB/ SILVACO TCAD software.

3. Text Books:

- T1:** “Modern V L S I Devices” Yuan Taur and Tak H Ning, Cambridge University Press, Fifth South Asia Edition 2017 Reprinted 2020.
T2: “Physics of Photonic Devices”, S. L. Chuang, Wiley Series in Pure and Applied Optics, 2009.

4. Reference Books:

- R1:** “Solid State Electronic Devices”, Streetman and Banerjee, PHI Learning Ltd., 2009.
R2: “Nanophotonics”, Paras N Prasad, Wiley Interscience, 2004..
R3: “Semiconductor Devices Physics and Technology”, S. M. Sze, John Wiley and Sons, INC Second Edition.

5. Course Plan:



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Module No.	Lecture Session	Reference	Learning outcomes
1	(L1-L2)Introduction and Overview	R2 Ch 1,2, Class	Overview of nano devices and integrated approach of electronics and photonics
2	(L3-L6)Semiconductor Fundamentals in Nanotechnology	TB1 Ch 2-3, Class	Fundamental of semiconductor material and charge transport characteristics
3	(L7-L10)Optical properties of nano-structures	TB2 Ch 7, R2 Chap.2, Class	Optical field confinement and field propagation through optical structures
4	(L11-L14)Scaling and Nano devices	TB1 Ch4, R1Ch.6, R3 Ch.6, Class	Issues and solutions in MOS scaling and its applications in nano devices and circuits
5	(L15-L19)Nano Electronic Material, Devices and circuits	R3 Ch. 11-13, Class	Design and development of electronic nano structures and devices
6	(L20-L24)High speed electronic devices and scaling	TB1 Ch 5 & 10, R3 Ch 7,Class	Concept of device performance enhancement and understanding of SOI MOSFETs, Hetero-structure, Strained, Resonant, Tunnel, and SET devices
7	(L25-L29) Integrated Optical circuits and Devices	TB1 (Ch 9), TB2 Ch 7-9, R2 Ch.9, Class notes	Concept of optical signal modification with fields, charge, energy and non-linear effects.
8	(L30-33) Optical sensors, devices and nonlinear operations	R1 Ch.8,R3 Ch.9 and class notes	Optical sources, detectors and circuits, nonlinear optical behaviour and its applications
9	(L34-L37)Nano photonic devices, circuits and systems	R2 Ch 14, Class	Nano optical structures and characteristics in communication, sensors, bio-sensor application
10	(L38-L40)New Trends in Nano device technology	Class	Research trends in nano electronics and photonics

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid Sem Test	90 mins	30	05/10/26 AN2	OB
Quizzes	5 mins	20		CB
Assignment/Seminar		15		
Compre	3 hrs	35	03/12/26 AN	OB/CB

7. Chamber Consultation Hour: Thursday 5.00 PM.

8: Home Assignment : These will be specified from time to time.

9. Notices : Nalanda



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10. Make-up Policy: Make-up will be given on extremely genuine grounds only. Prior application should be made for seeking the make-up examination.

11. Note (if any): Assignments and topics of general interest will be discussed in the class.

Instructor-in-charge
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